

Column densities

1 Installation

The actual column densities are calculated in Magritte, this can be cloned using

git clone --recursive <https://github.com/OwenVm/Magritte.git>

This is a fork of the main Magritte branch, with an extra functionality to calculate column densities quicker. Installation instructions can be found at https://magritte.readthedocs.io/en/stable/0_getting_started/2_installation.html

You might also need to install the latest version of plons, instructions can be found at https://plons.readthedocs.io/en/latest/0_getting_started/1_installation.html

2 Files

- `utils_raytracer.py`: This file contains the implementation of the ray-tracer in python.
- `cons.py`: This file I use to calculate the H_2 number density using partial pressures. If you change the way n_{H_2} is calculated, this is not needed anymore.
- `column_densities.py`: The main file that calculates the column densities. Make sure all the paths are set correctly. At the end, a file is made with a column density for every index. I also save the neighbour and boundary lists, such that these do not need to be recalculated if you want to rerun the code.
- `test.txt`: This is just a file that needs to be passed to Magritte in order to create a model, but this is not used at all.

3 Important parameters

There are two main parameters in `column_densities.py` that will influence the result (and computation time)

- **nside_interp:** (line 184) This determines the amount of rays that are traced in python, i.e., it will influence for how many points Magritte will calculate the column density. Some testing seems to suggest 2^{**3} suffices, corresponding to 768 rays (more should always be better, runtime scales roughly linear).
- **nrays:** (line 279) This determines in how many directions Magritte traces rays for each point. This parameter seems to be a bit more important than the **nside_interp** parameter. Computation time scales roughly linear. Little testing shows 192 suffices, and I would not go below this or you will start to see some artefacts of the ray tracer. (If you change this number, make sure it is of the form $12*x^{**2}$ with x some integer)

4 Limitation

Magritte needs a list of neighbours and boundary points for its ray tracer, hence the Delauney mesh is needed. It is possible that for some dump files this crashes, as I had it happen before for different models. Since I have only been testing the column densities calculations on the last dump, I am not sure whether this will happen for these models as well, anyway I expect it to work for most models. Also, the Delauney mesh calculation can take close to 50% of the time, as it runs on a single core. Creating the Magritte model also runs on a single core, and can hardly be sped up (this can also take a minute).